

CEN 481 Introduction to Data Mining

GETTING TO KNOW YOUR DATA -II

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Measuring Data Similarity and Dissimilarity

□ Similarity and dissimilarity measures are referred to as measures of **proximity**.

□ **Similarity**

- Numerical measure of how alike two data objects are
- Value is higher when objects are more alike
- It returns a value of 1 if the objects are identical. Often falls in the range $[0,1]$

□ **Dissimilarity** (e.g., distance)

- Numerical measure of how different two data objects are
- It works the opposite way. It returns a value of 0 if the objects are the same.
- The higher the dissimilarity value, the more dissimilar the two objects are.

□ Knowledge of object similarities can be used in clustering, outlier analysis, and nearest-neighbor classification.

Notation

- The objects are described by *multiple* attributes.
- Suppose that we have n objects described by p attributes:
 - $x_1 = (x_{11}, x_{12}, \dots, x_{1p}) \dots \rightarrow \dots x_n = (x_{n1}, x_{n2}, \dots, x_{np})$
 - x_{ij} means the value for object x_i of the j^{th} attribute

Data Matrix versus Dissimilarity Matrix

□ Data matrix (object-by-attribute structure)

- used to store the data objects
- n data points with p dimensions (n-by-p matrix)
- Two-mode matrix - made up of two kinds of entities (i.e. rows (for objects) and columns (for attributes))

$$\begin{bmatrix} x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ x_{n1} & \cdots & x_{nf} & \cdots & x_{np} \end{bmatrix}$$

□ Dissimilarity matrix (object-by-object structure)

- used to store dissimilarity values for pairs of objects
- n data points, but registers only the distance (n-by-n table). A triangular matrix...
- $d(i,j)$ is the measured **dissimilarity** or “difference” between objects i and j
- One-mode matrix - one kind of entity (dissimilarities)

$$\begin{bmatrix} 0 & & & & \\ d(2, 1) & 0 & & & \\ d(3, 1) & d(3, 2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n, 1) & d(n, 2) & \cdots & \cdots & 0 \end{bmatrix}$$

□ Measures of similarity can often be expressed as a function of measures of dissimilarity.

For example, for nominal data: $\text{sim}(i, j) = 1 - d(i, j)$

Proximity Measure for Nominal Attributes

- ❑ A nominal attribute can take on two or more states.
- ❑ Let the **number of states** of a nominal attribute be M .
- ❑ The states can be denoted by letters, symbols, or a set of integers, such as $1, 2, \dots, M$.
- ❑ Integers are used just for data handling and do not represent any specific ordering.

Proximity Measure for Nominal Attributes

- The dissimilarity between two objects i and j can be computed based on the ***ratio of mismatches***:

$$d(i, j) = \frac{p - m}{p}$$

where;

- where m is the number of matches (i.e., the number of attributes for which i and j are in the same state)
- p is the total number of attributes describing the objects

Note: *Weights can be assigned to increase the effect of m or to assign greater weight to the matches in attributes having a larger number of states*

Example

Proximity Measure for Nominal Attributes

□ Alternatively;

$$\text{sim}(i, j) = 1 - d(i, j) = \frac{m}{p}$$

□ Nominal attributes can be converted to binary attributes:

- For an object with a given state value, the binary attribute **representing** that state is set to **1**, while the **remaining** binary attributes are set to **0**.
- i.e. for attribute map_color => yellow attribute is set to 1, remaining to 0.

Proximity Measures for Binary Attributes

- ❑ Recall binary attributes have two states: 0 and 1
- ❑ Treating binary attributes as if they are numeric can be misleading.
- ❑ For computing dissimilarity compute a dissimilarity matrix from the given binary data.
 - If all binary attributes are thought of as having the same weight, we have the 2×2 contingency table.
 - The total number of attributes is: p , where $p = q + r + s + t$

Contingency Table for Binary Attributes

		Object j		
		1	0	
Object i	1	q	r	$q + r$
	0	s	t	$s + t$
	sum	$q + s$	$r + t$	p

Proximity Measures for Binary Attributes

- Dissimilarity that is based on symmetric binary attributes is called **symmetric binary dissimilarity**. For objects i and j described by symmetric binary attributes:

$$d(i, j) = \frac{r + s}{q + r + s + t}$$

- Dissimilarity based on asymmetric attributes is called **asymmetric binary dissimilarity**.

- the agreement of two 1s (a positive match) is considered more significant than that of two 0s (a negative match).
- the number of negative matches, t , is considered unimportant and is thus ignored:

$$d(i, j) = \frac{r + s}{q + r + s}$$

Proximity Measures for Binary Attributes

- We can also measure the difference between two binary attributes based on the notion of **similarity** instead of dissimilarity:
- For example, the **asymmetric binary similarity** between the objects i and j can be computed as:

$$\text{sim}(i, j) = \frac{q}{q + r + s} = 1 - d(i, j)$$

- The coefficient $\text{sim}(i, j)$ is called the **Jaccard coefficient**.

Example

Dissimilarity of Numeric Data

- ❑ **Euclidean**, **Manhattan**, and **Minkowski** distance measures are commonly used for computing the dissimilarity of objects described by numeric attributes.
- ❑ In some cases, the data are normalized before applying distance calculations.
 - i.e. transforming the data to fall within a smaller or common range, such as $[-1,1]$ or $[0.0,1.0]$.
 - Normalizing the data attempts to give all attributes an equal weight.

Dissimilarity of Numeric Data

□ **Euclidean distance**: Let $i=(x_{i1}, x_{i2}, \dots, x_{ip})$ and $j=(x_{j1}, x_{j2}, \dots, x_{jp})$ be two objects described by p numeric attributes. The Euclidean distance between objects i and j is defined as:

$$d(i, j) = \sqrt{(x_{i1} - x_{j1})^2 + (x_{i2} - x_{j2})^2 + \dots + (x_{ip} - x_{jp})^2}$$

□ **Manhattan (or city block) distance**: It is the distance in blocks between any two points in a city (such as 2 blocks down and 3 blocks over for a total of 5 blocks). It is defined as:

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

Dissimilarity of Numeric Data

□ Both the Euclidean and the Manhattan distance satisfy the following mathematical properties:

- **Non-negativity**: $d(i,j) \geq 0$: Distance is a non-negative number.
- **Identity of indiscernibles**: $d(i,i) = 0$: The distance of an object to itself is 0.
- **Symmetry**: $d(i,j) = d(j,i)$: Distance is a symmetric function.
- **Triangle inequality**: $d(i,j) \leq d(i,k) + d(k,j)$: Going directly from object i to object j in space is **no more than** making a detour over any other object k .

□ A measure that satisfies these conditions is known as **metric**.

Dissimilarity of Numeric Data

□ **Minkowski distance** is a generalization of the Euclidean and Manhattan distances. It is defined as:

$$d(i, j) = \sqrt[h]{|x_{i1} - x_{j1}|^h + |x_{i2} - x_{j2}|^h + \dots + |x_{ip} - x_{jp}|^h}$$

where h is a real number such that $h \geq 1$ (called **L_p norm**)

- It represents the Manhattan distance when $h=1$ (i.e., L_1 norm) and Euclidean distance when $h=2$ (i.e., L_2 norm).
- The **supremum distance** ($L_{\max}$, L_{∞} norm) is computed by finding the attribute f that gives the maximum difference in values between the two objects.

$$d(i, j) = \lim_{h \rightarrow \infty} \left(\sum_{f=1}^p |x_{if} - x_{jf}|^h \right)^{\frac{1}{h}} = \max_f |x_{if} - x_{jf}|.$$

Dissimilarity of Numeric Data

- If each attribute is assigned a weight according to its perceived importance, the **weighted Euclidean distance** can be computed as:

$$d(i, j) = \sqrt{w_1|x_{i1} - x_{j1}|^2 + w_2|x_{i2} - x_{j2}|^2 + \dots + w_m|x_{ip} - x_{jp}|^2}$$

- Weighting can also be applied to other distance measures as well.

Example

Proximity Measures for Ordinal Attributes

□ Suppose that f is an attribute from a set of ordinal attributes describing n objects. The **dissimilarity** computation with respect to ordinal attribute f :

1. The value of f for the i^{th} object is x_{if} , and f has M_f ordered states, representing the ranking $1, \dots, M_f$. Replace each x_{if} by its corresponding rank, $r_{if} \in \{1, \dots, M_f\}$.
2. Each ordinal attribute can have a different number of states, it is often necessary to map the range of each attribute onto $[0.0, 1.0]$ so that each attribute has equal weight. We perform such data normalization by replacing the rank r_{if} of the i^{th} object in the f^{th} attribute by:

$$z_{if} = \frac{r_{if} - 1}{M_f - 1}$$

3. Dissimilarity can then be computed using any of the distance measures, using z_{if} to represent the f value for the i^{th} object.

□ **Similarity** values for ordinal attributes can be interpreted from dissimilarity as $sim(i,j)=1-d(i,j)$.

Example

Dissimilarity for Attributes of Mixed Types

- ❑ In many real databases, objects are described by a mixture of attribute types.
- ❑ Combine the different attributes into a single dissimilarity matrix, bringing all of the meaningful attributes onto a common scale of the interval [0.0, 1.0].
- ❑ If the data set contains p attributes of mixed type, the dissimilarity $d(i, j)$ between objects i and j is defined as:

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

Dissimilarity for Attributes of Mixed Types

□ $\delta_{ij}^{(f)} = 0$ if

1. x_{if} or x_{jf} is missing (i.e., there is no measurement of attribute f for object i or object j)
2. $x_{if} = x_{jf} = 0$ and attribute f is asymmetric binary;

□ $\delta_{ij}^{(f)} = 1$ otherwise.

□ The contribution of attribute f to the dissimilarity between i and j (i.e., $d_{ij}^{(f)}$) is computed dependent on its type:

- If f is numeric: $d_{ij}^{(f)} = \frac{|x_{if} - x_{jf}|}{\max_h x_{hf} - \min_h x_{hf}}$ where h runs over all nonmissing objects for attribute f
- If f is nominal or binary: $d_{ij}^{(f)} = 0$ if $x_{if} = x_{jf}$; otherwise, $d_{ij}^{(f)} = 1$.
- If f is ordinal: compute the ranks r_{if} and $z_{if} = \frac{r_{if} - 1}{M_f - 1}$ and treat z_{if} as numeric.

$$d(i, j) = \frac{\sum_{f=1}^p \delta_{ij}^{(f)} d_{ij}^{(f)}}{\sum_{f=1}^p \delta_{ij}^{(f)}}$$

Example

Cosine Similarity

- ❑ Each document is an object represented by what is called a **term-frequency vector**.
- ❑ Term-frequency vectors are typically very long and sparse (i.e., they have many 0 values).
 - The traditional distance measures doesn't work since two term-frequency vectors may have many 0 values in common but this does not make them similar.

Document	team	coach	hockey	baseball	soccer	penalty	score	win	loss	season
<i>Document1</i>	5	0	3	0	2	0	0	2	0	0
<i>Document2</i>	3	0	2	0	1	1	0	1	0	1
<i>Document3</i>	0	7	0	2	1	0	0	3	0	0
<i>Document4</i>	0	1	0	0	1	2	2	0	3	0

Cosine Similarity

- ❑ *We need a measure that will focus on the words that the two documents do have in common, and the occurrence frequency of such words.*
- ❑ In other words, we need a measure for **numeric data** that ignores zero-matches.
- ❑ **Cosine similarity** is a measure of similarity that can be used to compare documents.

Cosine Similarity

□ Let x and y be two vectors for comparison:

$$\text{sim}(x, y) = \frac{x \cdot y}{||x|| ||y||}$$

where $||x||$ is the Euclidean norm of vector $x=(x_1, x_2, \dots, x_p)$, defined as:

$$\sqrt{x_1^2 + x_2^2 + \dots + x_p^2}$$

It is the length of the vector. Similarly, $||y||$ is the Euclidean norm of vector y .

□ The measure computes the cosine of the angle between vectors x and y .

- A **cosine value of 0** means that the two vectors are at 90 degrees to each other (orthogonal) and **have no match**.
- The closer the cosine value to 1, the smaller the angle and **the greater the match between vectors**.

Example